Canteen Automation System

A Project Report

Submitted by

Ms. Divya Manik Rathod
(Roll no. 202037)

Fulfillment for the award of the degree

BACHELOR OF COMPUTER APPLICATIONS

UNDER THE GUIDANCE OF
PROF. NAZIYA BANEDER

MKSSS'S
K.B. JOSHI INSTITUTE OF INFORMATION TECHNOLOGY,
KARVE NAGAR, PUNE 52

AFFILIATED BY
SNDT WOMEN'S UNIVERSITY, MUMBAI.
2022-2023

MAHARSHI KARVE STREE SHIKSHAN SAMSTHA'S
(Approved By Govt. of Maharashtra, NAAC Accredited)
Approved by AICTE New Delhi, DTE (Dte Code – 6971),
Affiliated to S.N.D.T. Women's University, Mumbai

K.B. JOSHI INSTITUTE
OF INFORMATION
TECHNOLOGY
KARVE NAGAR, PUNE 411052

CERTIFICATE

This is to certify that the project report titled

"Canteen Automation System"

Miss Divya Rathod **Roll no. 202037**

Is a bonafide work carried out by them under the supervision of Mrs. Naziya Banedar. Project is completed and approved for fulfillment of the requirement SNDT Women's University, Mumbai forth award of Degree of Bachelor of Computer Application (BCA).

The project work was carried during the period 2022-23

Project Guide **Principal**

External Examiner **Internal Examiner**

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Phone- 020-25477599,020-25470171 Email : kbjiitbca@maharshikarvebcapune.org

Institution Exclusively for Women's Education

Maharshi Karve Stree Shikshan Samstha

महर्षी कर्वे स्त्री शिक्षण संस्था

120 Years Journey for women's Empowerment through Education

"शीलं परं भूषणम्"

Certificate

Date:

This is to certify that Miss Divya Rathod and has satisfactorily completed the project titled "Canteen Automation System" application under my guidance/ supervision for my Canteen work. Both of them have Analyzed our requirements properly and developed the working product within the stipulated time.

Mr. Girish Kumthekar

(Manager)

INDEX

1) Acknowledgement
2) Introduction
3) Company Profile
4) Existing System
5) Drawbacks of Existing System
6) Proposed System
7) Objective of Proposed System
8) Scope of Proposed System
9) Feasibility Study
10) Hardware and Software Requirements for Proposed System
11) Context Level Diagram
12) 1st Level Diagram (DFD)
13) Sequence Diagram
14) Use Case Diagram
15) Entity Relationship Diagram (ERD)
16) File Designing
17) Input Screen
18) Testing and Implementation Planning
19) Bibliography

ACKNOWLEDGEMENT

We have a great pleasure and satisfaction in presenting the project on “Canteen Auto machine System”. We would like to show our gratitude towards our Principal Dr. Swati Sayankar for providing good facilities and valuable guidance.

We would like to express our gratitude to Prof. Naziya Banedar for their project guidance and support, there guidance throughout helped us in completion of project.

We would like to thank Mr. Girish kumthekar for showing faith in us and allowing us to develop software “Canteen Automation System” for their Canteen.

We are thankful for all teaching, non-teaching staff of the college who directly and indirectly helped us in completion of project.

INTRODUCTION

Computers have become part of the life for accessing almost any kind of information. In today's age of fast food and take-out, many canteens have chosen to focus on quick preparation and speedy delivery of orders rather than offering a rich dining experience. What, we propose is a Canteen Automation System, which is a technique of ordering foods online applicable in any food delivery industry.

The main advantage of this system is that it greatly simplifies the ordering process for both the customer and the canteens. When the customer visits the ordering webpage, they are presented with an interactive and up-to-date menu, complete with all available options and dynamically adjusting prices based on the selected options. After making a selection, the item is then added to their order, which the customer can review the details of at any time before checking out.

This provides instant visual confirmation of what was selected and ensures that items in the order are, in fact, what was intended.

This system also greatly lightens the load on the canteen's end, as the entire process of taking orders is automated. Once an order is placed on the webpage, it is entered into the database and then retrieved, in pretty much real-time, by a desktop application on the canteen's end. Within this application, all items in the order are displayed, along with their corresponding options and delivery details, in a concise and easy to read manner. This allows canteen employees to quickly go through the orders as they a are placed and produce the necessary items with minimal delay and confusion.

COMPANY PROFILE

- **COMPANY NAME**
  Suswaad Bakery

- **OWNER NAME**
  Girish Kumthekar

- **ADDRESS**
  MKSSS'S Campus, Karve Nagar, Pune 411052

- **CONTACT NO.**
  9225521401

EXISTING SYSTEM

- The existing system is a cash and paper-based system. The payment and process take a lot of time as the customer has to pay the exact amount and wait for the change.

- If the change is not available at the time, a coupon is provided which should be shown at the counter at the next purchase.

DRAWBACKS OF EXISTING SYSTEM

- The existing system is much time consuming.
- Much inefficient as the chances of mistakes are increased in manual work.
- It is difficult to manage numerous registers.
- It is unsafe as anyone can have access to it.

PROPOSED SYSTEM

- Canteen Automation System simplify everything from customer order, product management.

- Web based system that can be access anywhere makes daily task easier, simpler, and faster for everyone

- System automatic will be generated either by cash or online payment.

PROJECT OBJECTIVE

- To build an automated system.
- To order food rapidly.
- To provide complete information about canteen dishes along with their rates and description.
- To maintain customer records in an easy way.
- Easy to add, edit, update, or delete any canteen and dish.
- To make it convenient for people who have limited time.
- Computerized order.

SCOPE OF PROPOSED SYSTEM

- Our system will maintain an automated Dish handling facility which controls all the manual's function.
- It will provide search facilities for quick items, customer.
- Provide necessary reports to the customer as well as administrator of System.
- Customer order can be handled easily.
- It will be easy to create new item and add item to the available admin side with all the details.
- Customer payment will be excepted either by cash or online payment.
- Restricted to access the unauthorized users.
- Facility to change the available product details.

FEASIBILITY STUDY

A feasibility study for a blood bank management system on Java would involve examining the technical, economic, and operational aspects of the project to determine whether it is viable and worthwhile to pursue. Here are some of the factors that would be considered in such a study:

Technical Feasibility:

- Is HTML & PHP a suitable language for the project
- It Are the necessary resources available, such as servers and databases, to support the system?
- Can the system be integrated with existing blood bank equipment and software?

Economic Feasibility:

- What is the estimated cost of development, deployment, and maintenance of the system?
- Will the benefits of the system, such as increased efficiency and accuracy, outweigh the costs?
- What is the return on investment (ROI) for the system?

Operational Feasibility:

- Will the system be easy to use and understand for blood bank personnel?
- How will the system impact the workflow of the blood bank?
- Will the system be able to handle the volume of blood donations and requests?

Hardware & Software Requirements

Hardware Requirement:

1.i3 Processor Based Computer or higher
2. Memory: 4GB RAM
3. Hard Disk Capacity: 50 GB
4. Internet Connection

Software Requirement:

1. Windows & or higher
2. Visual Studio 2010
3. SQL Server 2008

Language:

HTML, CSS, PHP

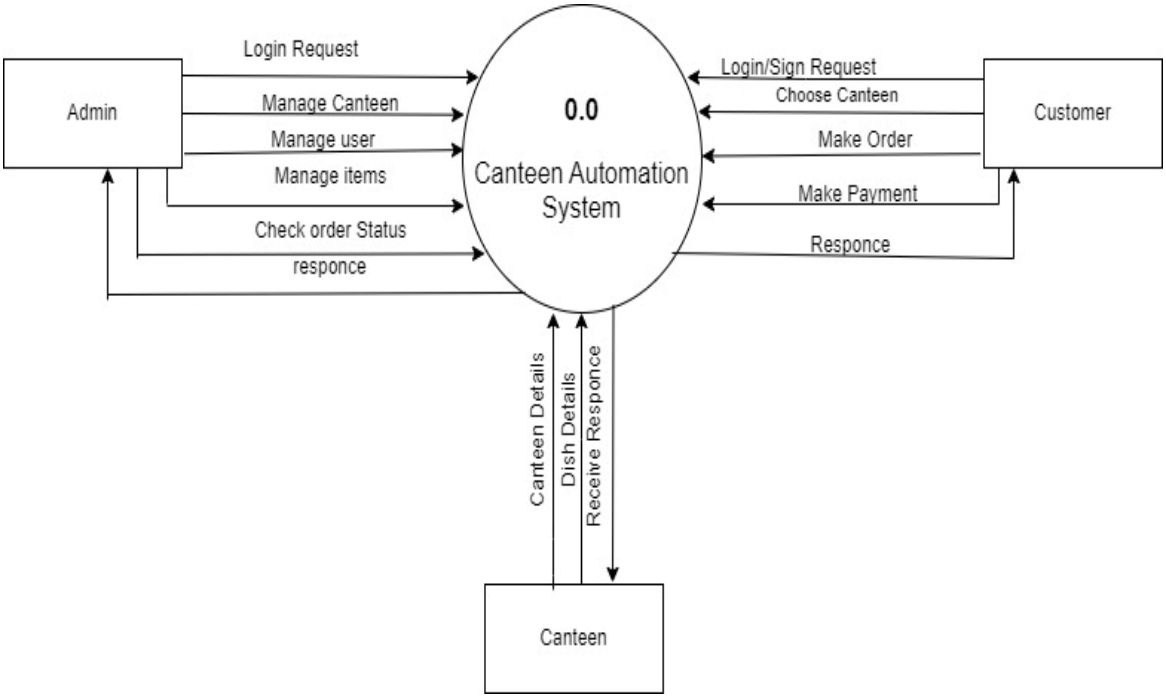
Database:

MySQL

DIAGRAM

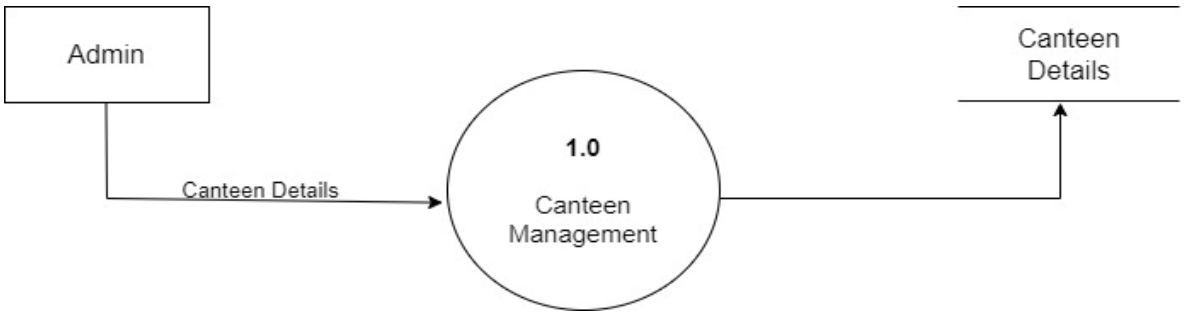
.

CLD (Context Level Diagram)

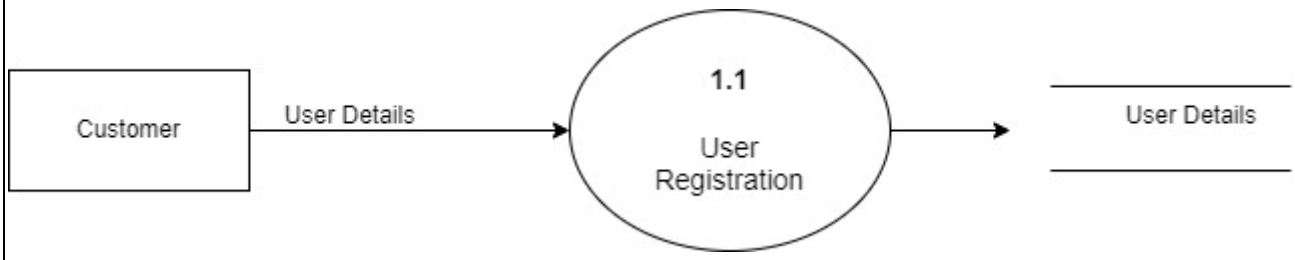


DFD (Data Flow Diagram)

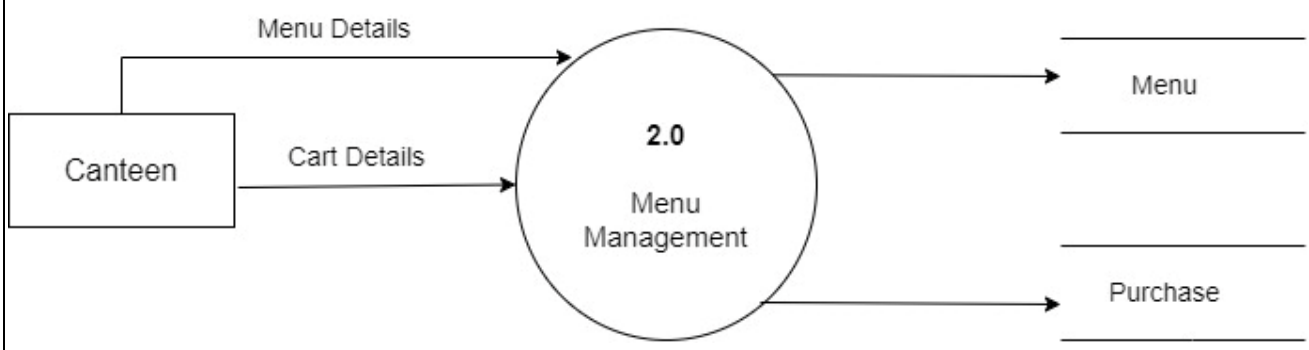
Canteen Information Gathering:



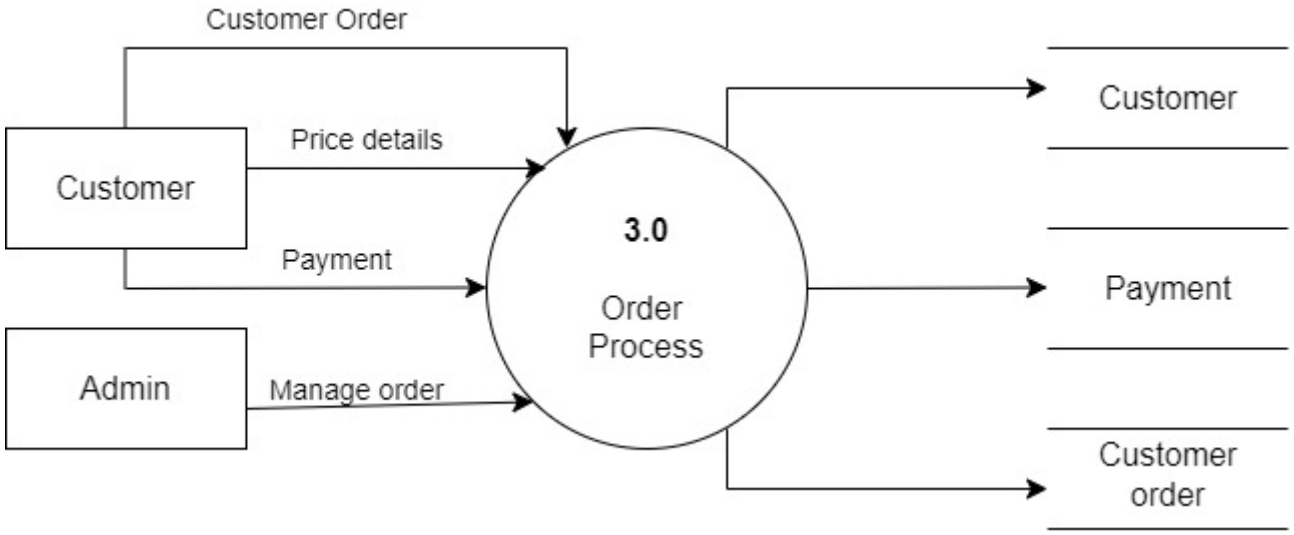
User Registration:



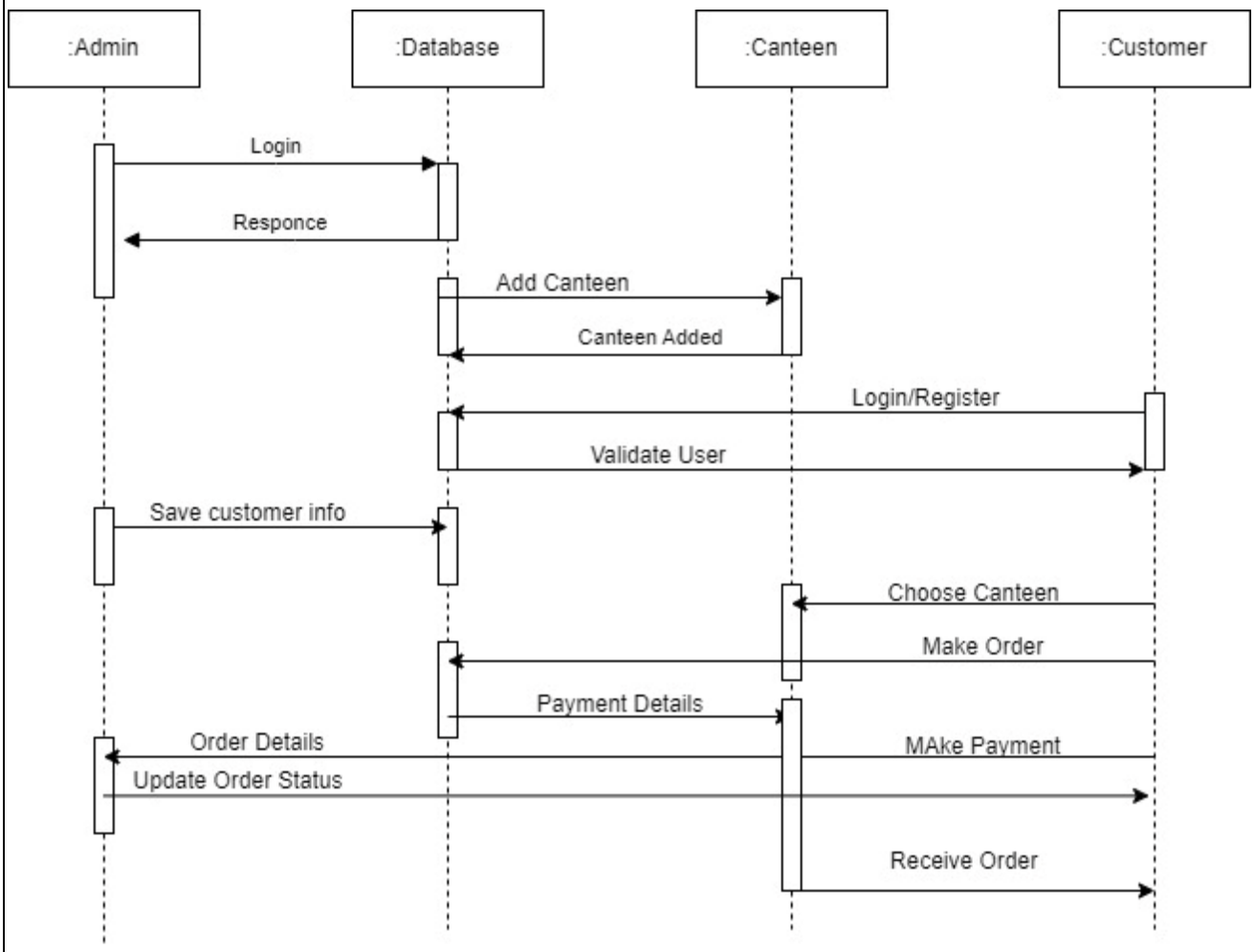
Menu Management Details:



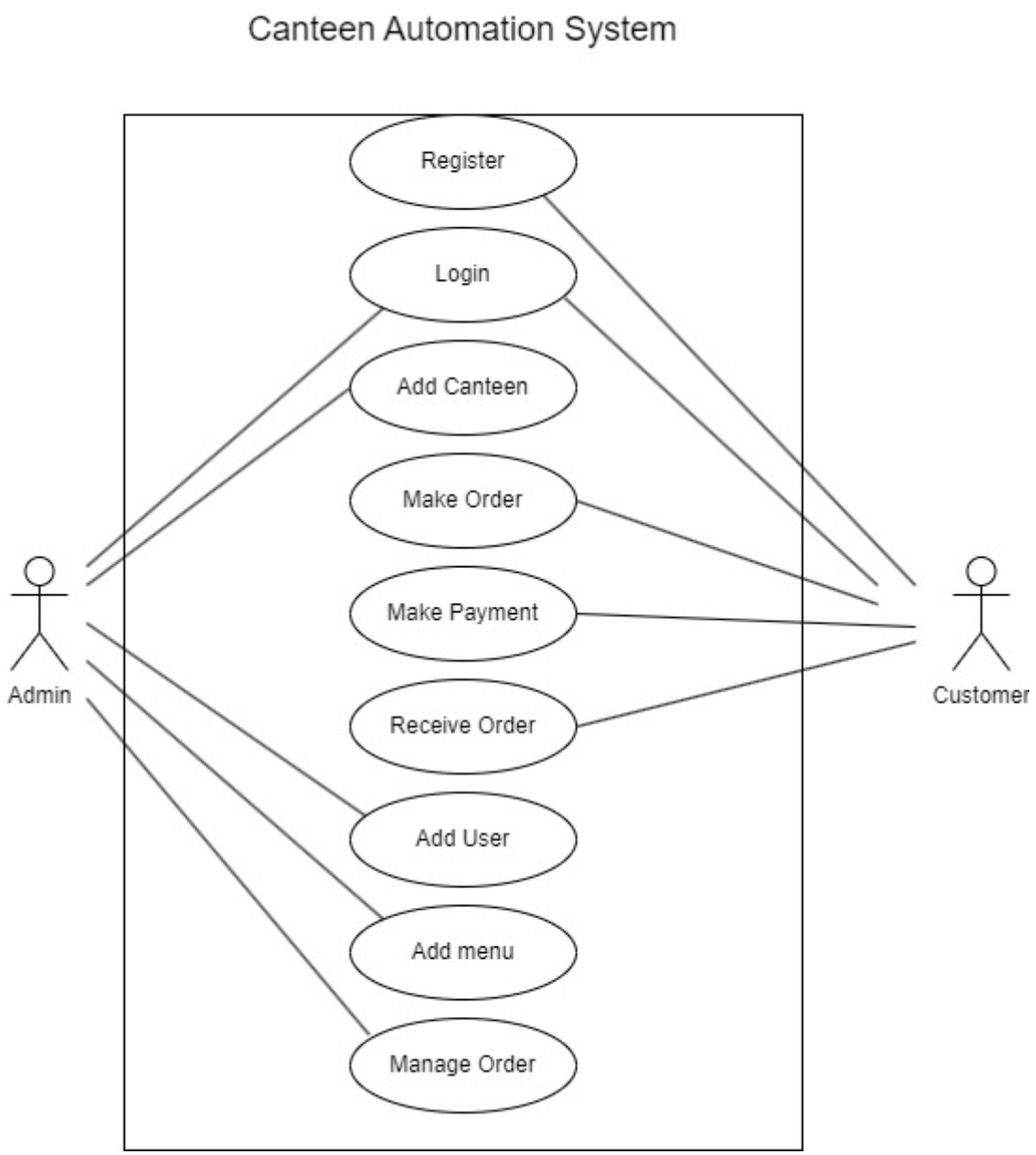
Order Process:



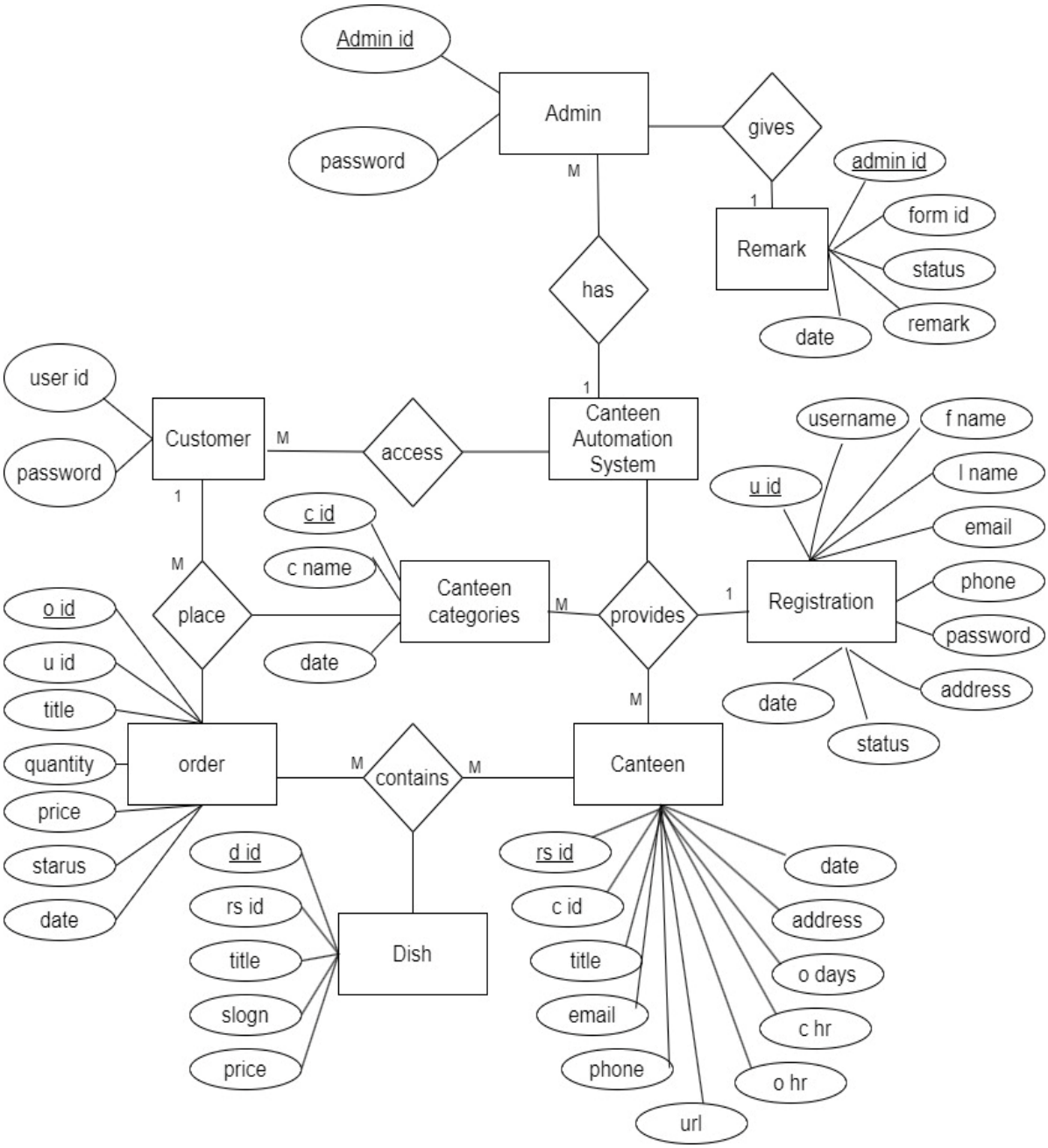
Sequence Diagram



Use Case Diagram



ERD (Entity Relationship Diagram)



FILE DESIGNING

Admin

Name Field	Data Type	Size	Description	Constraints
ADMIN_ID	INT	50	ADMIN ID	PRIMARY KEY
USERNAME	VARCHAR	50	ADMIN NAME	NOT NULL
PASSWORD	VARCHAR	50	PASSWORD	NOT NULL
EMAIL	VARCHAR	10	EMAIL	NOT NULL
CODE	VARCHAR	10	ADMIN CODE	NOT NULL
DATE	TIMESTAMP	10	DATE	NOT NULL

Admin Code

Name Field	Data Type	Size	Description	Constraints
ID	INT	50	ADMIN ID	PRIMARY KEY
CODE	VARCHAR	6	ADMIN CODE	NOT NULL

Restaurant Category

Name Field	Data Type	Size	Description	Constraints
C_ID	INT	50	CATEGORY_ID	PRIMARY KEY
C_NAME	VARCHAR	6	CATEGORY_NAME	NOT NULL
DATE	VARCHAR		DATE	NOT NULL

Restaurant

Name Field	Data Type	Size	Description	Constraints
RS_ID	INT	50	RESTAURANT ID	PRIMARY KEY
C_ID	INT	50	CATEGORY ID	FOREIGN KEY
TITLE	VARCHAR	50	TITLE	NOT NULL
EMAIL	VARCHAR	10	EMAIL	NOT NULL
PHONE	VARCHAR	10	PHONE NUMBER	NOT NULL
URL	TIMESTAMP	10	WEBSITE	NOT NULL
O_HR	VARCHAR	222	OPENING HOUR	NOT NULL
C_HR	VARCHAR	222	CLOSING HOUR	NOT NULL
O_DAYS	VARCHAR	222	OPENING HOUR	NOT NULL
ADDRESS	TEXT		ADDRESS	NOT NULL
IMAGE	TEXT		IMAGE	NOT NULL
DATE	TIMESTAMP		DATE	NOT NULL

Dishes

Name Field	Data Type	Size	Description	Constraints
D_ID	INT	50	DISH ID	PRIMARY KEY
RS_ID	INT	50	RESTAURANT NAME	FOREIGN KEY
TITLE	VARCHAR	50	DISH TITLE	NOT NULL
SLOGAN	VARCHAR	222	SLOGAN OF DISH	NOT NULL
PRICE	DECIMAL	10,2	DISH PRICE	NOT NULL
IMG	VARCHAR	10	DISH IMG	NOT NULL

USER

Name Field	Data Type	Size	Description	Constraints
U_ID	INT	50	USER ID	PRIMARY KEY
USERNAME	VARCHAR	50	USER NAME	NOT NULL
F_NAME	VARCHAR	50	FIRST NAME	NOT NULL
L_NAME	VARCHAR	10	LAST NAME	NOT NULL
EMAIL	VARCHAR	10	EMAIL	NOT NULL
PHONE	VARCHAR	10	PHONE	NOT NULL
PASSWORD	VARCHAR	222	PASSWORD	NOT NULL
ADDRESS	TEXT	222	ADDRESS	NOT NULL
STATUS	INT	222	STATUS	NOT NULL
DATE	TIMESTAMP		DATE	NOT NULL

USER_ORDER

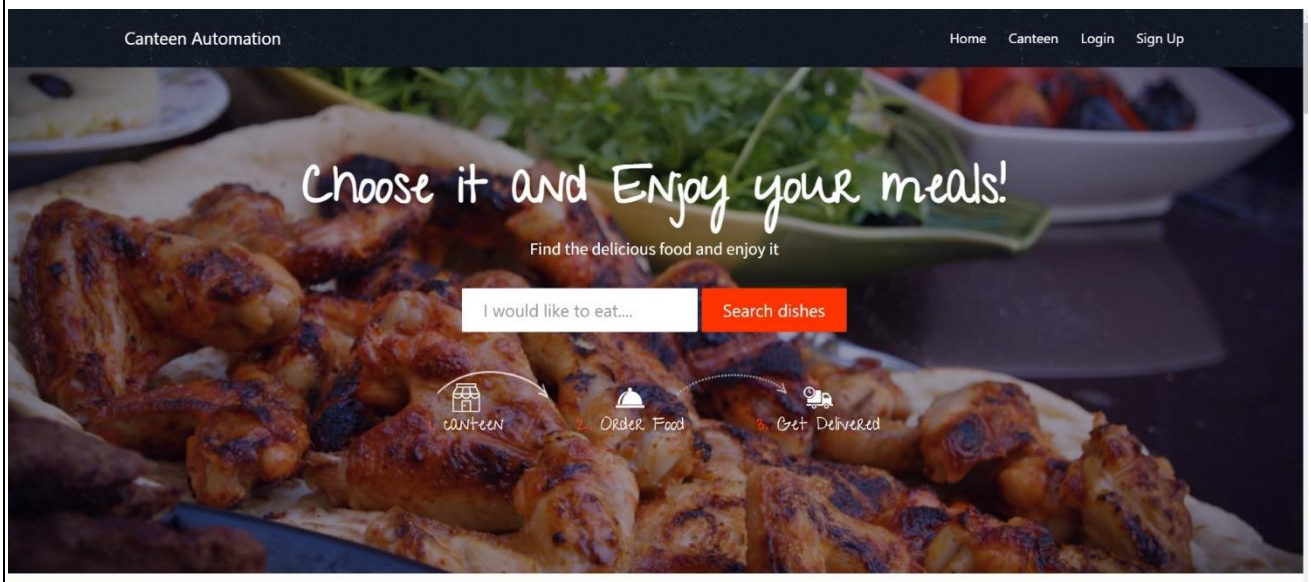
Name Field	Data Type	Size	Description	Constraints
O_ID	INT	50	USER ID	PRIMARY KEY
U_ID	INT	50	USER NAME	FOREIGN KEY
TITLE	VARCHAR	50	DISH TITLE	NOT NULL
QUANTITY	VARCHAR	10	QUANTITY	NOT NULL
PRICE	DECIMAL	10,2	PRICE	NOT NULL
STATUS	VARCHAR	222	STATUS	NOT NULL
DATE	TIMESTAMP		DATE	NOT NULL

Remark

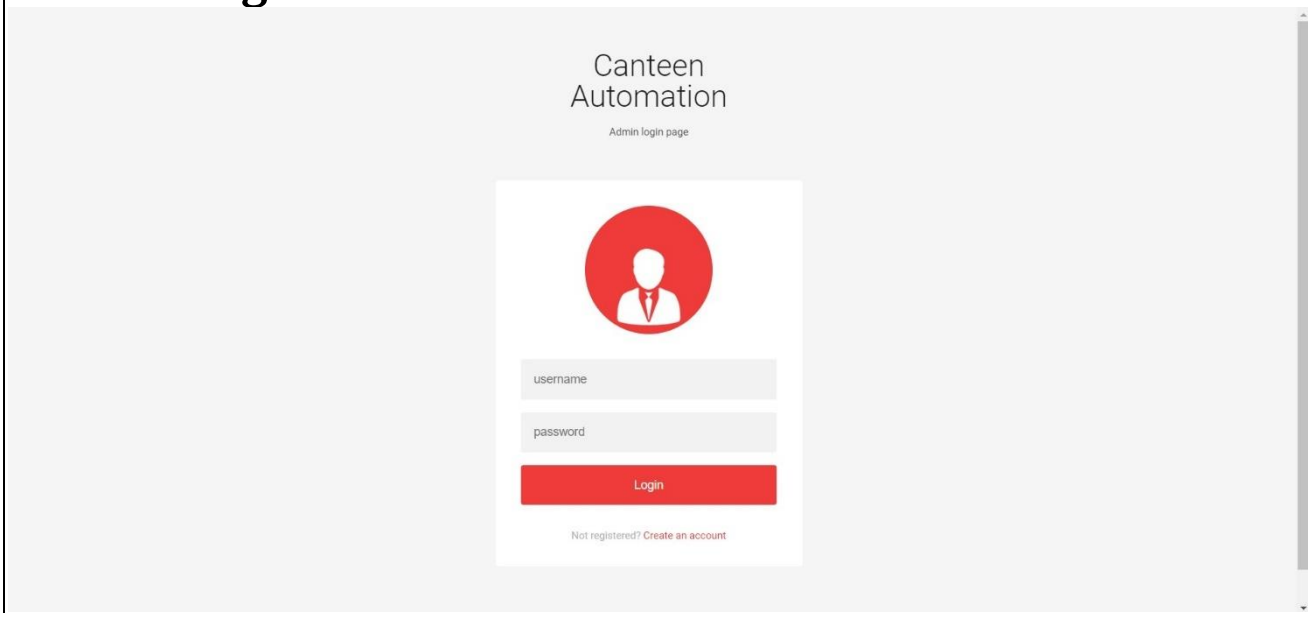
Name Field	Data Type	Size	Description	Constraints
ID	INT	50	USER ID	FOREIGN KEY
FRM_ID	INT	50	USER NAME	NOT NULL
STATUS	VARCHAR	255	DISH TITLE	NOT NULL
REMARK	MEDIUMTEXT		QUANTITY	NOT NULL
REMARKDATE	TIMESTAMP		PRICE	NOT NULL

Input Screens

Home Page:



Admin Login:



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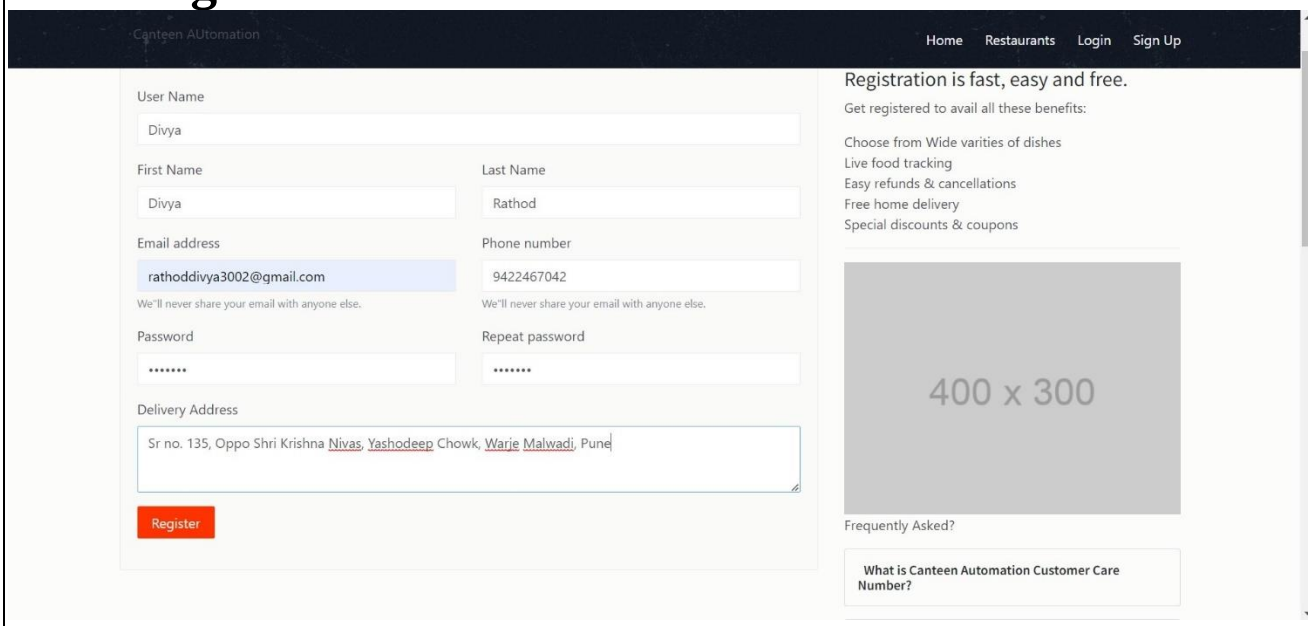

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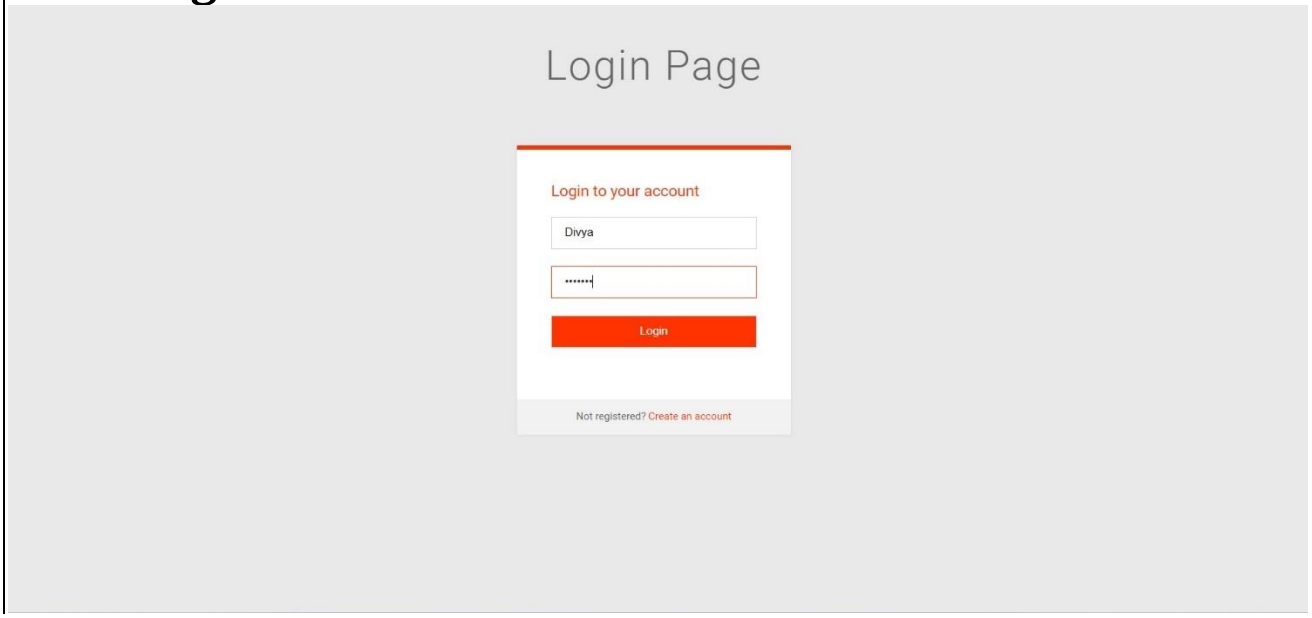
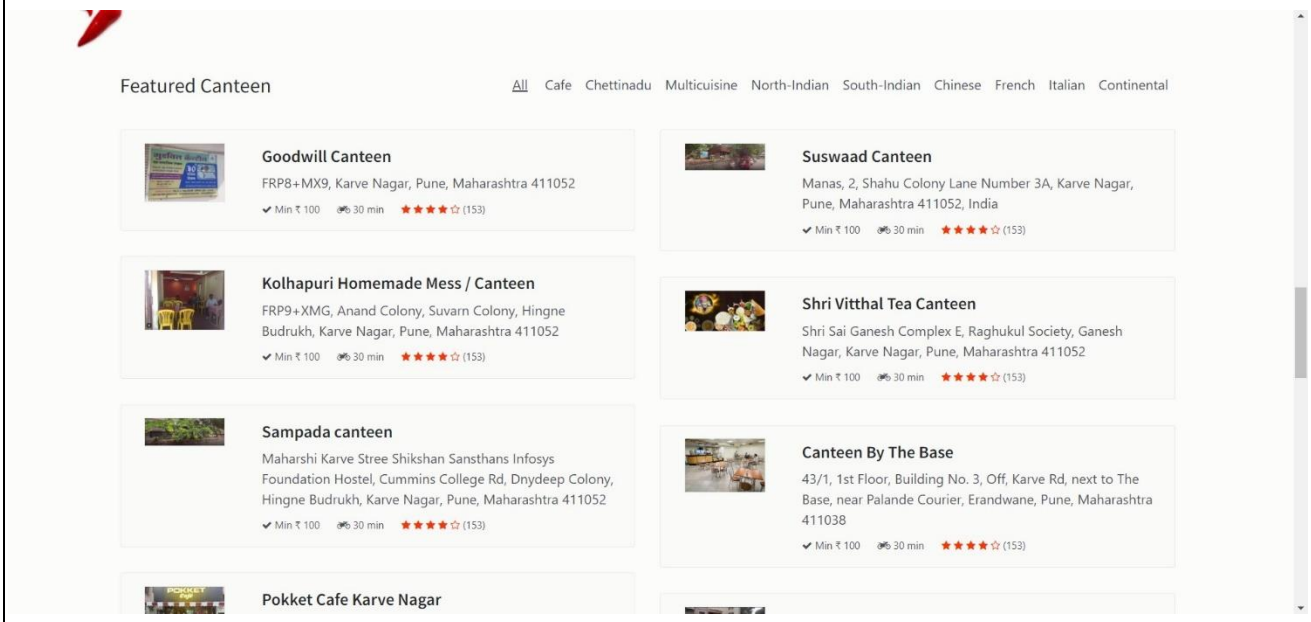
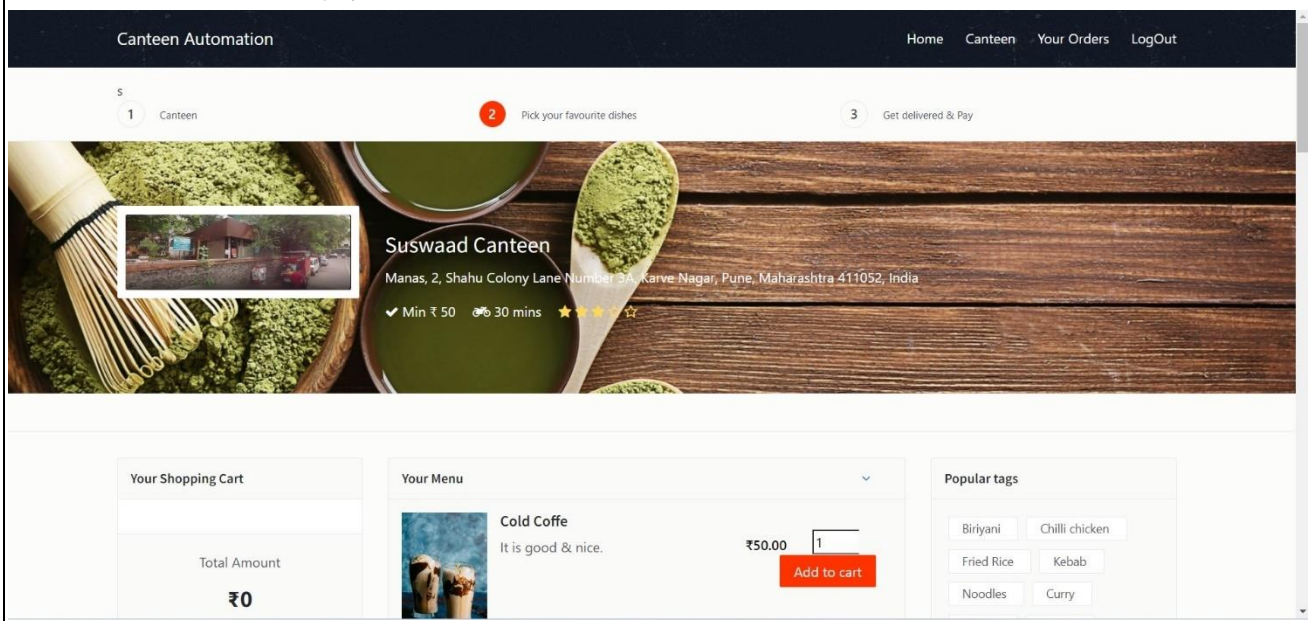


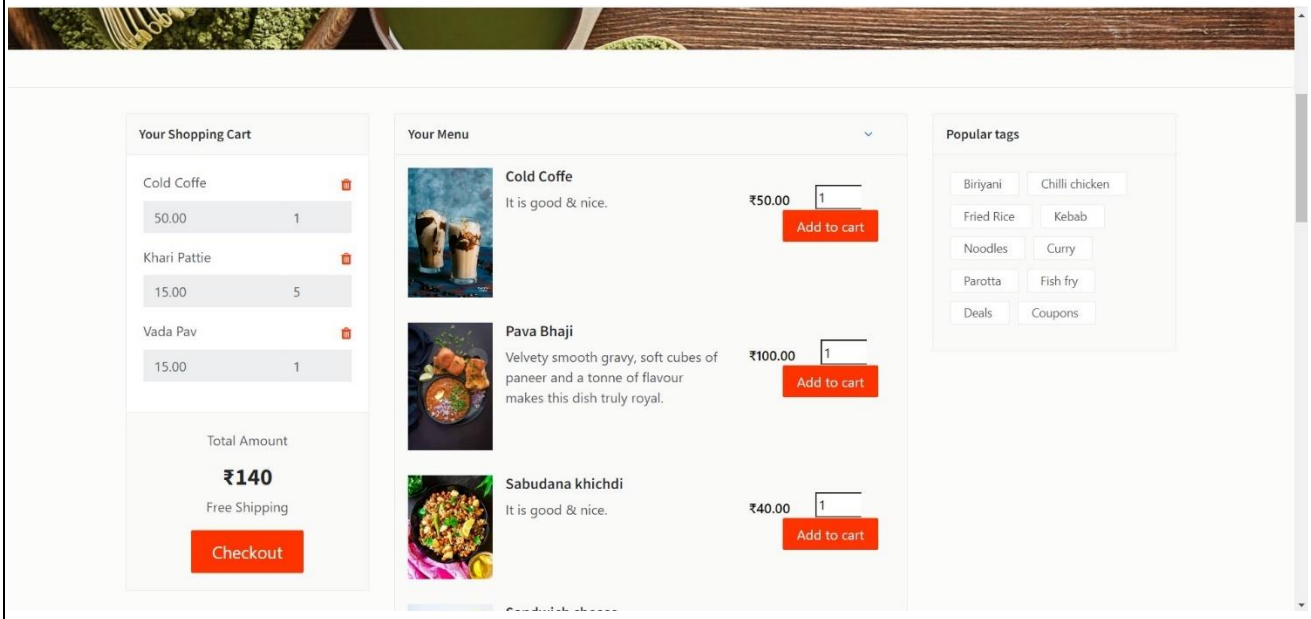
List:



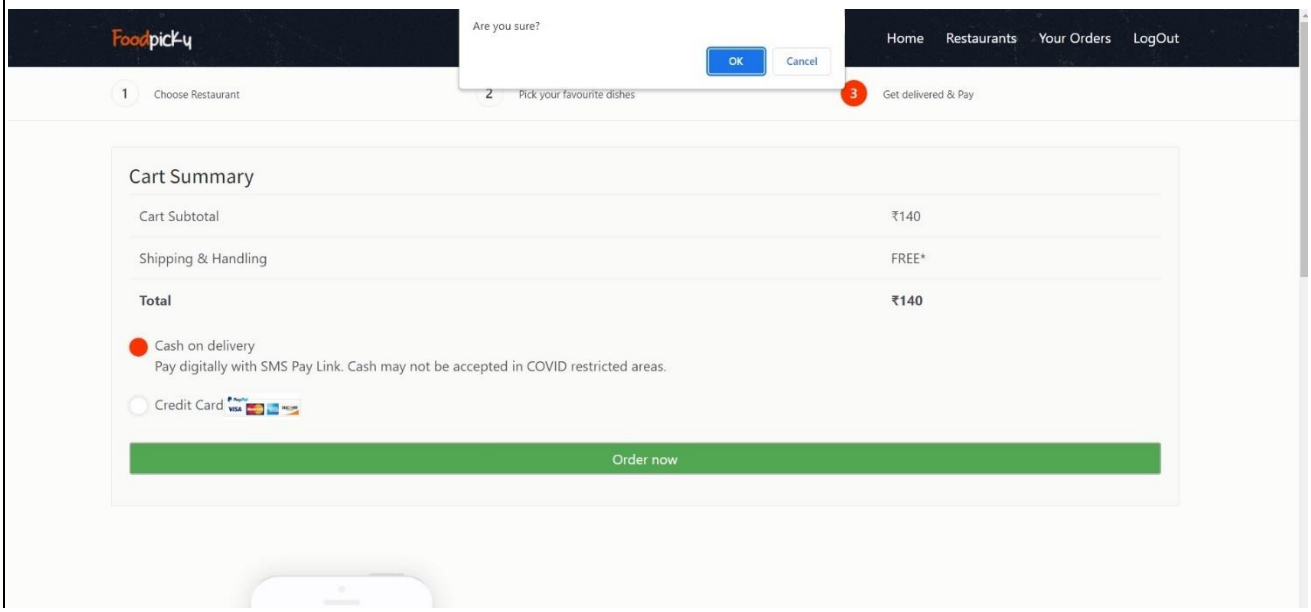
Canteen Menu:



Checkout:



Order Confirmation:



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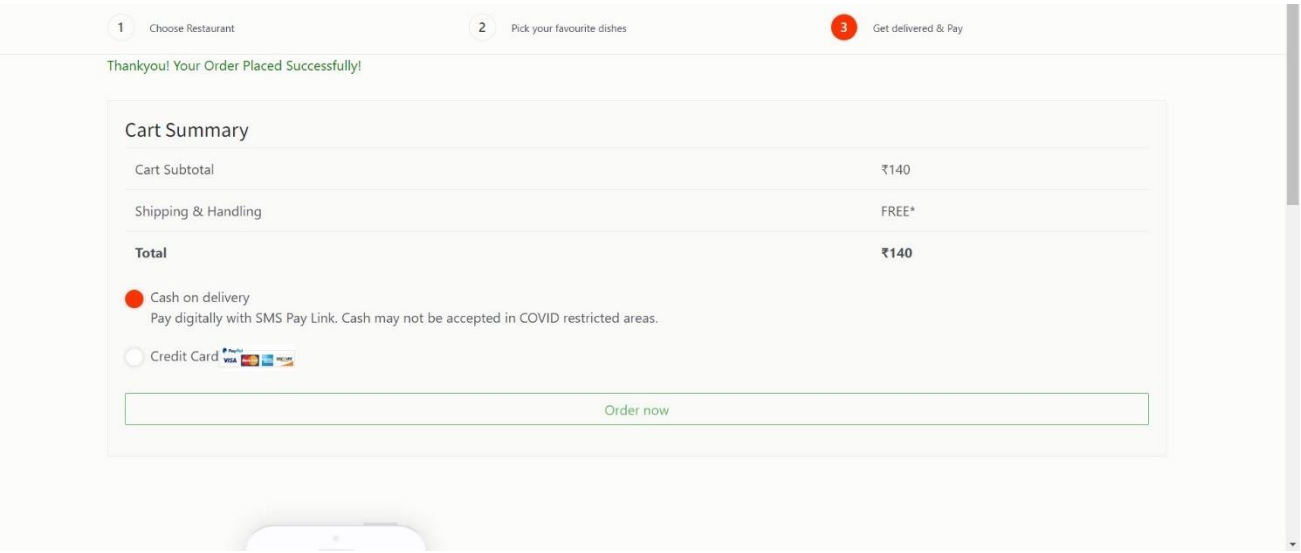
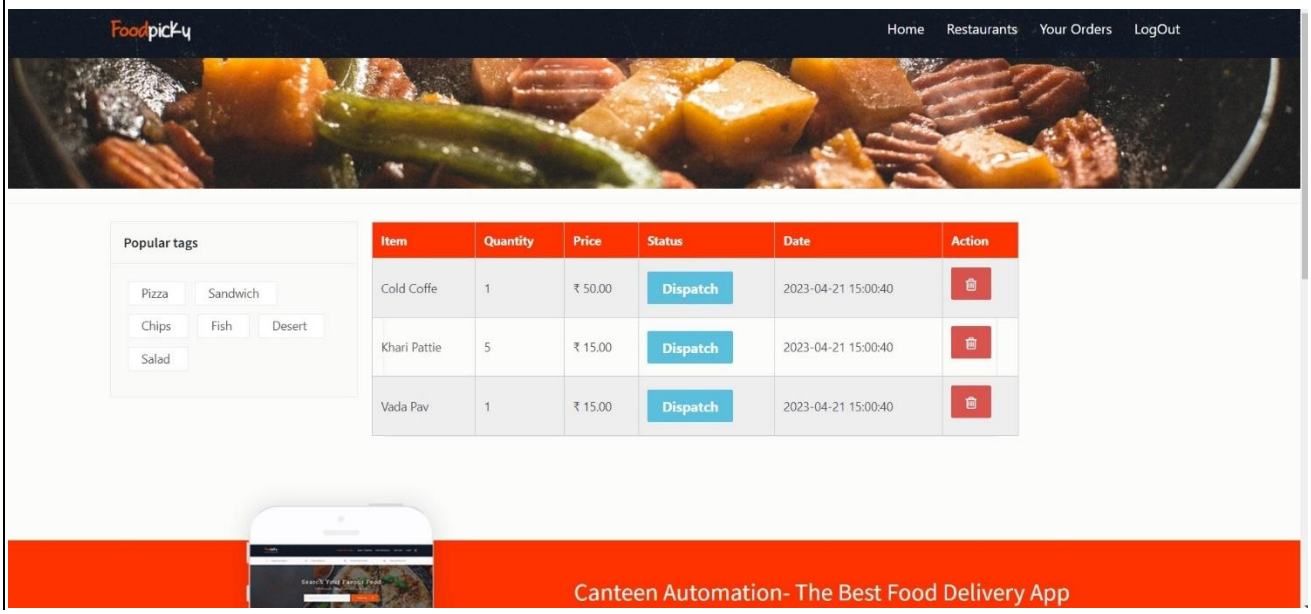
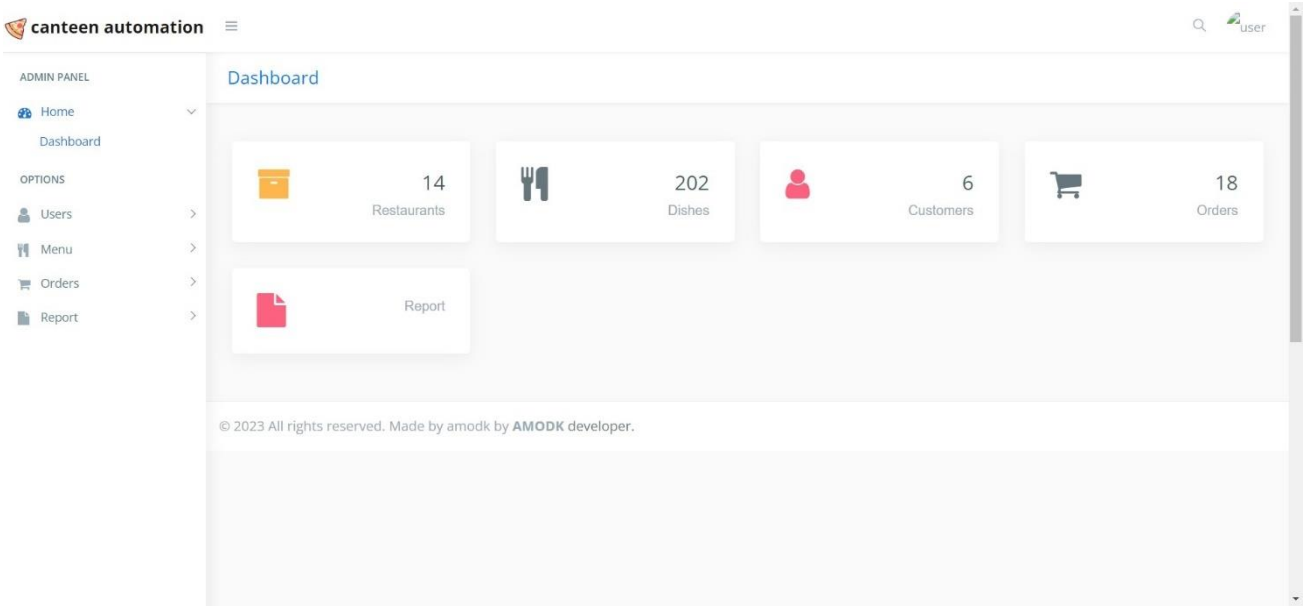


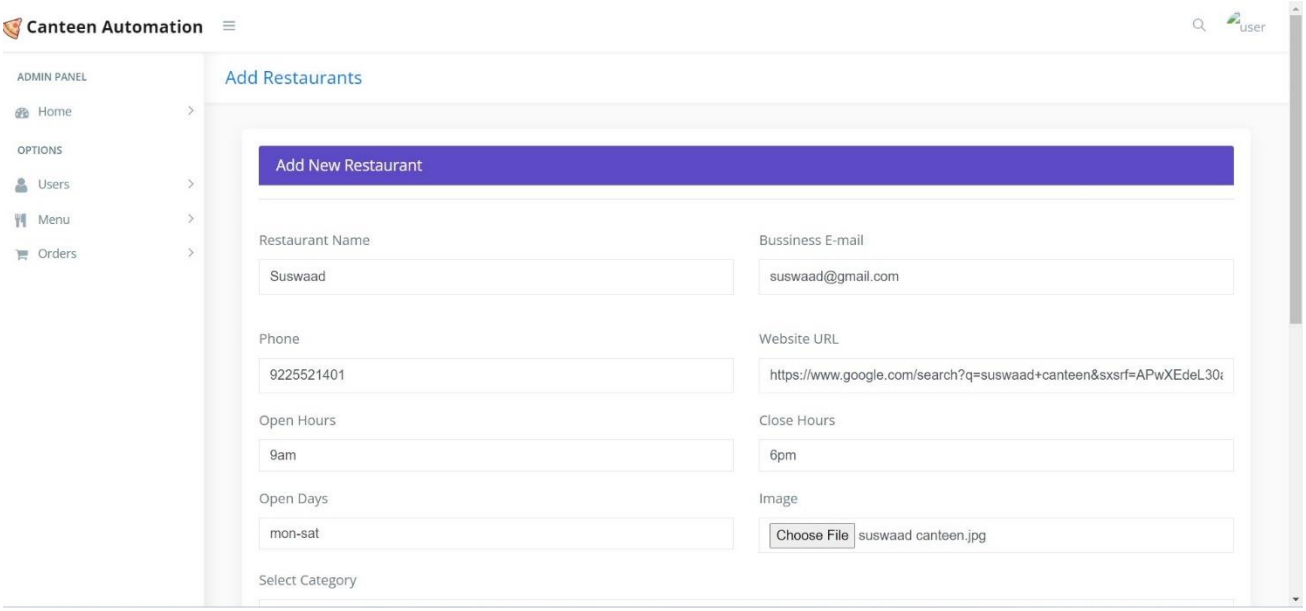
User Order:



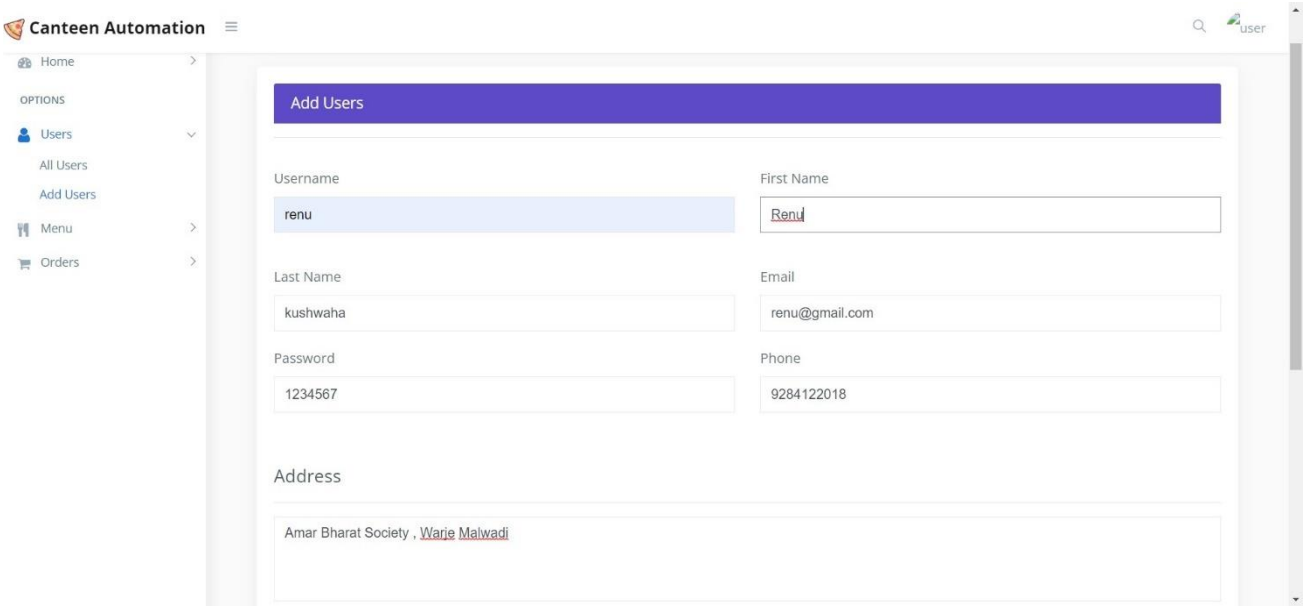
Admin Dashboard:



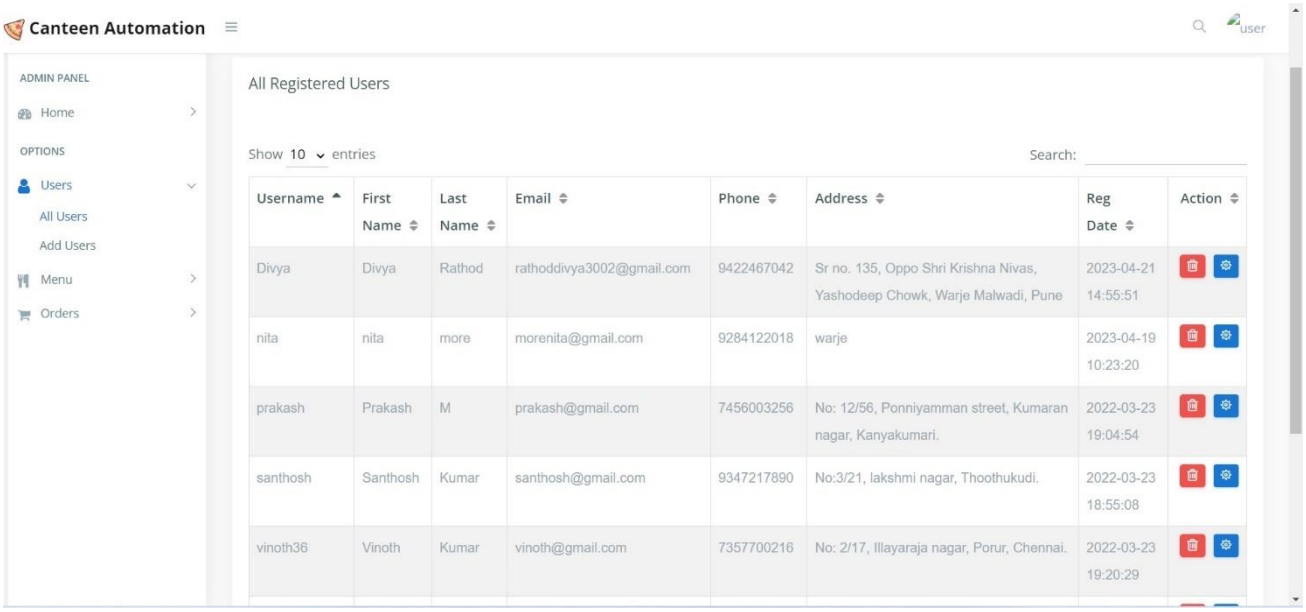
Add Canteen:



Add User:

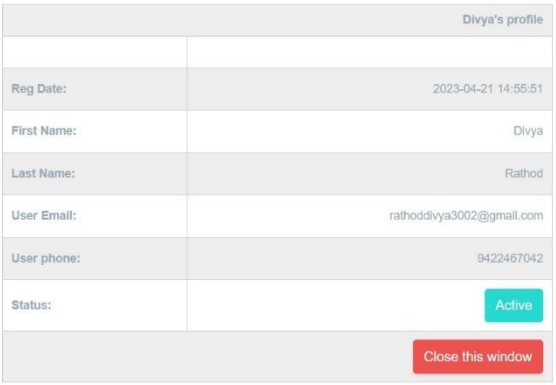


All User:



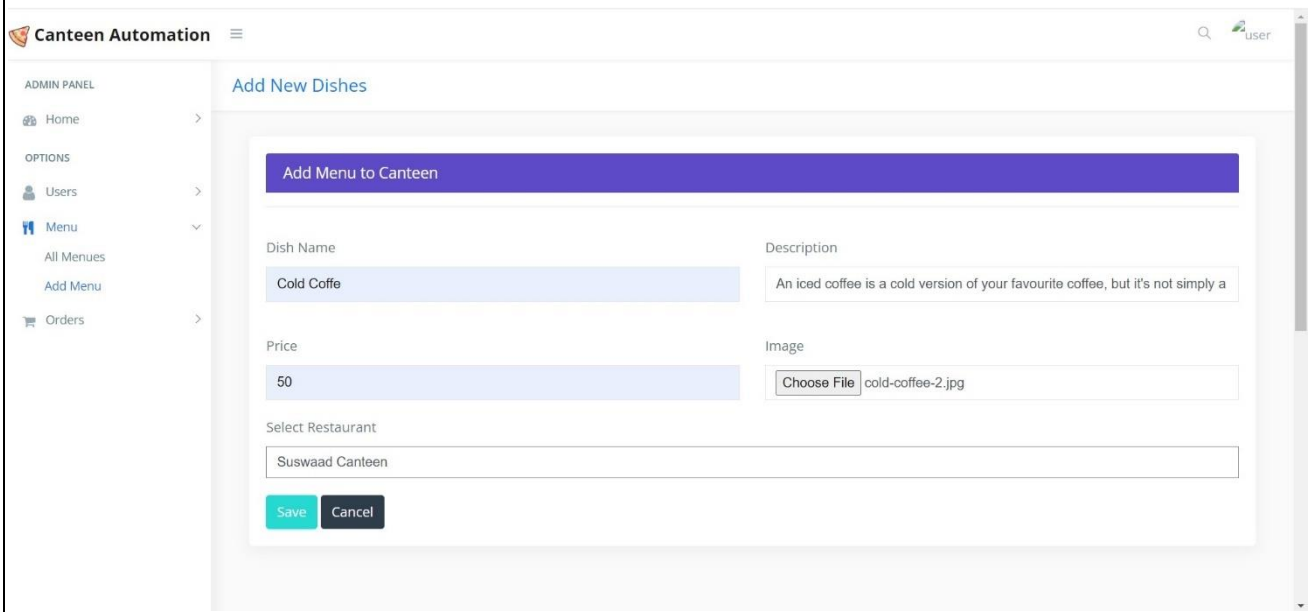
Username	First Name	Last Name	Email	Phone	Address	Reg Date	Action
Divya	Divya	Rathod	rathoddivya3002@gmail.com	9422467042	Sr no. 135, Oppo Shri Krishna Nivas, Yashodeep Chowk, Warje Malwadi, Pune	2023-04-21 14:55:51	
nita	nita	more	morenita@gmail.com	9284122018	warje	2023-04-19 10:23:20	
prakash	Prakash	M	prakash@gmail.com	7456003256	No: 12/56, Ponniyamman street, Kumaran nagar, Kanyakumari.	2022-03-23 19:04:54	
santhosh	Santhosh	Kumar	santhosh@gmail.com	9347217890	No:3/21, lakshmi nagar, Thoothukudi.	2022-03-23 18:55:08	
vinoth36	Vinoth	Kumar	vinoth@gmail.com	7357700216	No: 2/17, Illayaraja nagar, Porur, Chennai.	2022-03-23 19:20:29	

View User Details:

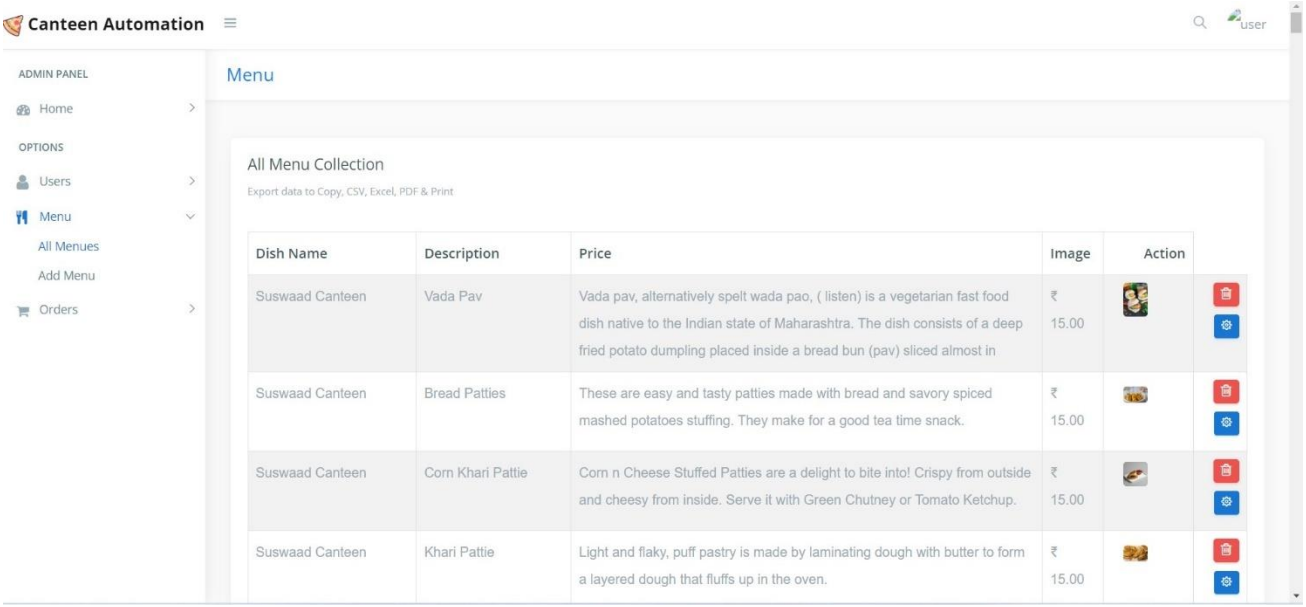


Divya's profile	
Reg Date:	2023-04-21 14:55:51
First Name:	Divya
Last Name:	Rathod
User Email:	rathoddivya3002@gmail.com
User phone:	9422467042
Status:	Active
	Close this window

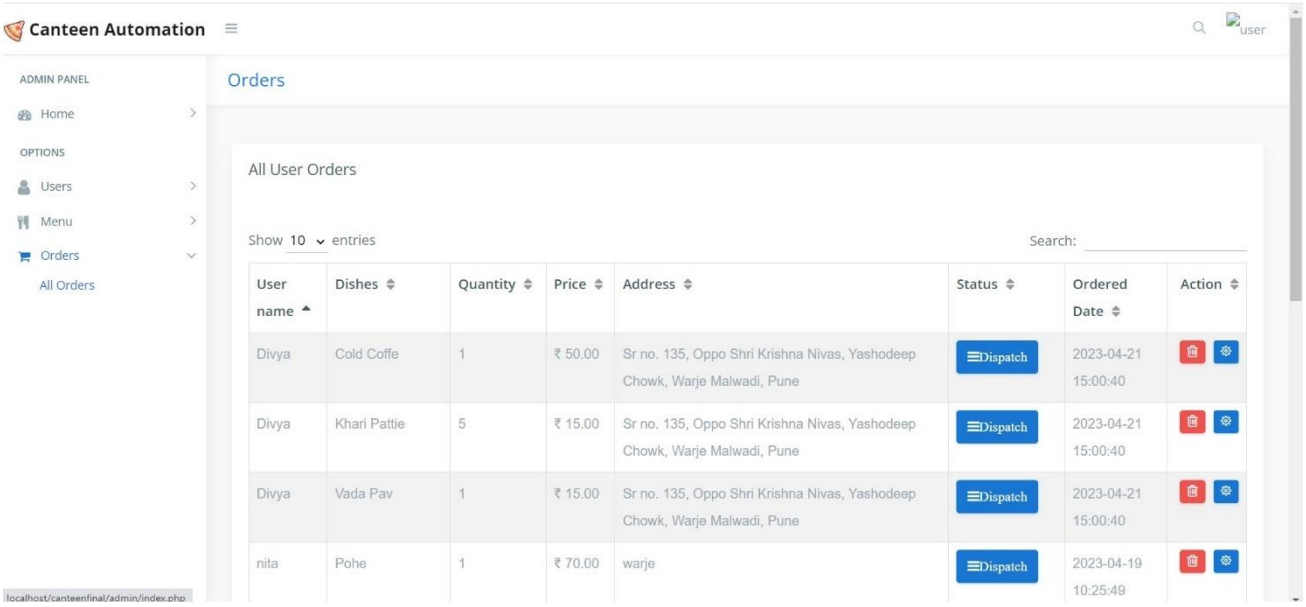
Add Menu:



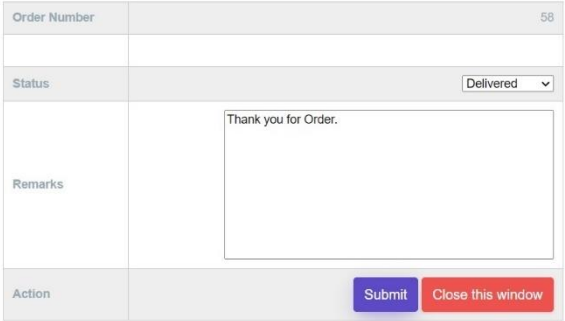
All Menu:



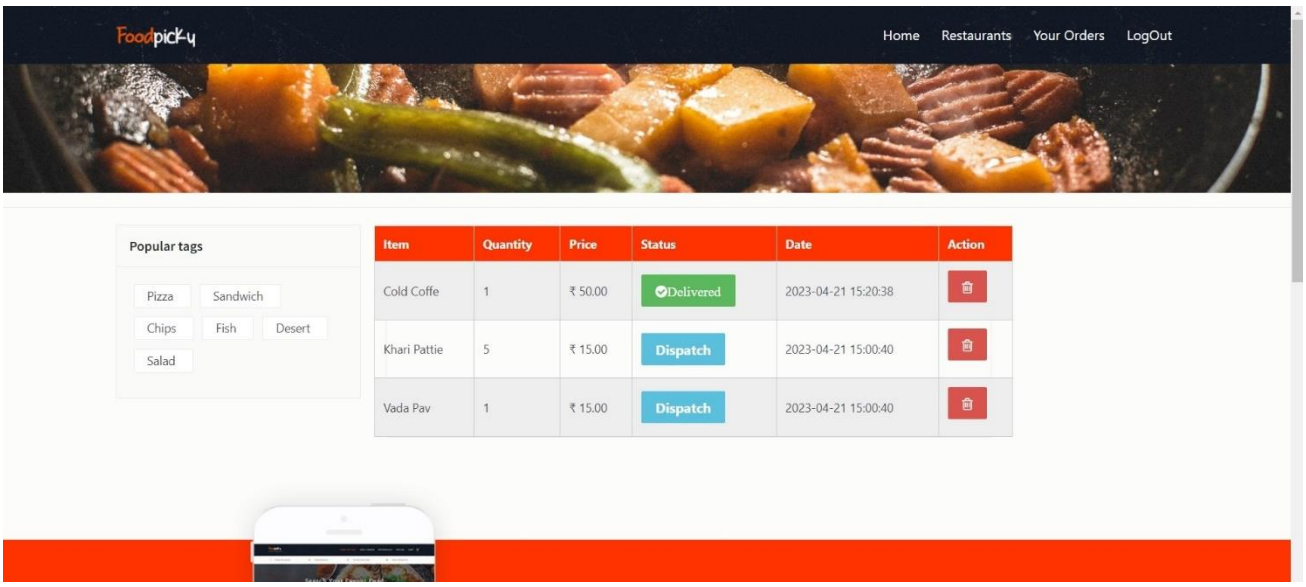
All Orders:



Order Status:



Delivered Order:



Report:

Monthly Report

Month	Revenue	Expenses	Profit
January	RS.1000	Rs.800	Rs.200
February	Rs.1200	RS.900	Rs.300
March	Rs.1500	Rs.1000	Rs.500
April	RS.1100	Rs.700	Rs.100
MAY	RS.2100	Rs.900	Rs.300

Testing and Implementations

Through testing and implementations, the system was carried out in a systematic approach. In the testing stage different levels of testing take place so as to ensure that the system is free failure.
In testing process different levels of testing have been used as mentioned below.
1. Unit testing.
2. Integration question.

The overall testing process was accomplished through the following steps:

1. Unit testing
Unit testing was done to focus on the individual program module. Here the goal was to assure that each program module function properly as a module.
White box testing technique was used to accomplish the unit testing. When the satisfied output is achieved, the further testing data takes place step by step.

2. Integration testing
After unit testing the number of models is assembled together to form a complete package and then the total package and top-down approach was allowed for an integration testing of the system. Black Book testing technique was used to accomplish the unit testing.

Bibliography

https://www.w3schools.com/
https://www.geeksforgeeks.org/
www.javatpoint.com